AI-Powered Energy-Efficient Storage for Renewable Grids with Predictive Analytics and IoT-Enabled BMS

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I. ABSTRACT

The integration of renewable energy sources into modern power grids introduces challenges such as intermittency, voltage instability, and limited efficiency in conventional storage systems. This project presents an AI-powered energy-efficient storage system for renewable grids, built around a 12 V, 9 Ah lithium-ion battery pack integrated with an IoT-enabled Battery Management System (BMS) using an ESP32 microcontroller [3]. The system implements real-time thermal monitoring using J-type thermocouples embedded in copper busbars, combined with physics-informed neural network (PINN) based thermal prediction models for early hotspot detection and active thermal management through PWM-controlled 12 V DC cooling fans [3]. Advanced AI techniques, including hybrid GRU-based deep learning models with ridge regression preprocessing, enable accurate State-of-Charge (SOC) and State-of-Health (SOH) estimation, achieving approximately 98% SOC accuracy and over 93% SOH prediction accuracy across simulated and experimental cycles [1,2,4]. A 6-pulse MOSFET-based three-phase inverter, enhanced with an LC harmonic filter, performs efficient DC–AC conversion from 12 V DC to 415 V AC, supporting both grid-tied and islanded operating modes with switching latency below 50 ms. Additionally, a reinforcement learning-based Proximal Policy Optimization (PPO) energy dispatch controller optimizes power flow among solar PV, grid supply, and battery storage, achieving over 91% dispatch efficiency and reducing grid downtime by approximately 38% during renewable fluctuations. IoT connectivity using MQTT communication and Astra DB cloud integration enables remote monitoring, predictive maintenance, and scalable deployment for microgrid and smart energy applications [4,5,6].

II. INTRODUCTION

a. Background

The global transition toward renewable energy sources such as solar photovoltaic (PV) and wind power is accelerating to mitigate climate change and reduce reliance on fossil fuels. According to the International Energy Agency (IEA), renewables are expected to contribute nearly 50% of global electricity generation by 2030. However, their intermittent and variable nature introduces operational challenges, including voltage fluctuations, frequency deviations, and energy curtailment, which can reach 20–30% in developing grids. Lithium-ion (Li-ion) batteries have emerged as key energy storage solutions due to their high energy density (up to 250 Wh/kg) and long cycle life (>1000 cycles). Yet conventional systems often suffer from limited thermal monitoring, inaccurate SOC/SOH estimation, and static energy management, leading to safety risks such as thermal runaway, which accounts for ~15% of battery failures, and inefficient utilization under dynamic loads [3]. To address these gaps, this project proposes an AI-powered energy-efficient storage system integrating embedded sensing, predictive analytics, and IoT connectivity, improving reliability, safety, and operational efficiency in renewable grid applications.

b. Motivation

The motivation arises from the need for reliable microgrid solutions, especially in rural regions like India, where grid instability affects over 300 million people and renewable penetration is below 20%. Traditional BMS approaches, relying on Coulomb counting, accumulate errors up to 10%, while passive cooling fails to detect localized thermal hotspots, increasing degradation and failure risk [3]. Energy dispatch between solar PV, battery, and grid is often rule-based, resulting in 15–20% energy wastage. By integrating AI for SOC/SOH prediction, reinforcement learning for intelligent energy dispatch, and IoT cloud monitoring, the proposed system improves operational awareness, adaptive control, and battery lifespan by ~20% [1,2,4].

c. Objectives

The project aims to design a scalable Li-ion storage unit with: (1) an integrated BMS for overcharge/over-discharge protection and cell balancing, (2) cell-level thermal fault detection with <0.5 s response, (3) AI-driven SOC/SOH estimation using GRU-based models with >90% accuracy, (4) a grid-synchronized inverter with LC-filtered three-phase AC output, and (5) reinforcement learning-based energy dispatch

to optimize efficiency and reduce downtime. Collectively, these objectives demonstrate a smart, autonomous, and efficient energy storage system suitable for microgrid, rural electrification, and renewable integration applications [1,2,4,5,6].

III. LITERATURE SURVEY

a. State-of-Health and State-of-Charge Estimation in Li-ion Batteries

Journal Paper: IEEE - *Transformer-Based Deep Learning Strategies for Lithium-Ion Batteries SOX Estimation Using Regular and Inverted Embedding*

Accurate estimation of State-of-Charge (SOC) and State-of-Health (SOH) is crucial for battery longevity and safe operation in renewable systems. Traditional methods like Coulomb counting accumulate errors over cycles due to unmodeled nonlinearities in degradation. Recent advancements leverage data-driven AI approaches to address this. For instance, a Transformer-based deep learning framework has been proposed for SOC and SOH prediction, utilizing attention mechanisms to capture temporal dependencies in voltage-current profiles, achieving mean absolute errors below 2% on NASA datasets. Similarly, multi-input Long Short-Term Memory (LSTM) networks have demonstrated robust SOH estimation by processing sequential charge-discharge data, outperforming support vector machines with 95% accuracy in real-time scenarios. Another study integrates machine learning with external parameters like temperature and impedance for electric vehicle (EV) batteries, reducing estimation variance to 3% under varying loads, highlighting the potential of hybrid models for grid storage. These works inspire our hybrid LSTM-GRU model, which synchronizes cycles for online prediction, enhancing adaptability to renewable intermittency.

b. Thermal Management and Fault Detection in Battery Packs

Journal Paper: IEEE - *Thermal Fault Diagnosis Method for Lithium-ion Batteries Based on Electrochemical Impedance Spectroscopy*

Thermal faults, such as internal short circuits, pose severe risks in Li-ion packs, often leading to thermal runaway with propagation rates up to 1 m/s. Conventional passive cooling via airflow reduces surface temperatures but overlooks localized heating at cell tabs. Physics-informed machine learning has emerged as a solution, with one method using Distribution of Relaxation Times (DRT) analysis to diagnose thermal faults via characteristic peak detection in impedance spectra, enabling early isolation with 98% precision. For vehicular packs, soft short-circuit diagnosis employs model-based observers and Kalman filters, detecting anomalies in voltage deviations >0.1 V within 100 ms, thus preventing cascade failures. Comparative AI techniques further validate sensor fusion for fault localization, integrating thermal and electrical models to achieve <5% false positives in multi-cell configurations. Our system builds on these by embedding thermocouples in busbars for hotspot mapping, augmented with active PID cooling to sustain uniform temperatures.

c. Inverter Designs for Renewable Energy Integration

Journal Paper: IEEE - *Design of a low cost DC/AC inverter for integration of renewable energy sources into the smart grid*

Efficient power conversion is essential for injecting renewable DC into AC grids. Low-cost DC/AC inverters have been designed for smart grid tie-ins, employing SPWM modulation to minimize harmonics (<3% THD) while handling 1-5 kW loads from PV sources. A 15-level cascaded H-bridge inverter enhances solar PV efficiency to 96%, reducing switching losses through selective harmonic elimination and phase-shift control. For large-scale wind integration, grid-forming inverters with virtual synchronous machine emulation ensure stability in weak grids (short-circuit ratio <3), supporting black-start capabilities. These designs inform our 6-pulse MOSFET inverter, which uses optocoupler isolation and PLL synchronization for bidirectional flow, tailored to 415 V 3-phase outputs.

d. AI-Based Energy Dispatch in Renewable Grids

Journal Paper: IEEE - *Artificial Intelligence-Based Methods for Economic Power Dispatch in Smart Grids*

Intelligent dispatch optimizes resource allocation amid variability. AI techniques, including genetic algorithms and neural networks, have been reviewed for economic dispatch in smart grids, yielding 10-15% cost reductions by forecasting demand and renewables output. An AI-driven EMS for solar microgrids employs fuzzy logic and particle swarm optimization to prioritize PV-battery charging, achieving 90% self-sufficiency in off-grid modes. Generative adversarial networks (GANs) further enable predictive optimization, simulating scenarios to enhance reliability by 25% in hybrid systems. Drawing from these, our reinforcement learning agent rewards efficiency and stability, dynamically switching sources to minimize losses in fluctuating conditions.

IV. METHODS AND MATERIALS

This section details the methodologies and materials employed in the development of the AI-Powered Energy-Efficient Storage System for renewable grids. The project adopts a hybrid engineering approach that synergistically combines hardware prototyping for energy storage and conversion, embedded sensing for real-time monitoring, and AI-driven algorithms for predictive analytics and intelligent energy dispatch. The hardware core is a 12V, 9 Ah lithium-ion battery pack equipped with modular safety mechanisms, while the software stack integrates machine learning models for SOC/SOH estimation, predictive fault detection, and reinforcement learning-based energy management. Calibration thresholds, such as thermal alerts at $\Delta T > 5°C/min$, and simulation benchmarks, including 200 synthetic charge–discharge cycles for AI model training, ensure that the system is scalable and suitable for microgrid and smart grid applications.

a. Hardware Components

The hardware stack is designed for modularity, cost-effectiveness (total BOM ~₹12,000), and robustness under variable renewable inputs. Key subsystems include the battery pack, sensing network, inverter, and auxiliary controls, assembled using off-the-shelf components from suppliers like Digi-Key and local vendors.

The hardware design prioritizes modularity, cost-effectiveness (~₹12,000 BOM), and robustness against variable renewable energy inputs. The primary subsystems include the battery pack, thermal and electrical sensing network, inverter, and auxiliary controls. The battery pack is a 3p3s configuration of 18650 Li-ion cells, yielding 12V, 9 Ah, and 216Wh energy capacity, with a custom BMS board rated at 10A. The BMS integrates overcharge/over-discharge cutoffs, short-circuit protection via MOSFETs (IRF540N), and cell balancing circuits. Cells are interconnected using nickel strips and copper busbars (10×2×0.5 mm) for uniform current distribution and thermal conduction, housed in a 4×4 acrylic enclosure with thermal pads, rated IP54.

Thermal sensing is performed with four J-type thermocouples embedded in the busbars, interfaced via MAX6675 amplifiers, enabling cell-level hotspot detection with ±1.5°C accuracy. Active cooling is implemented with PWM-controlled 12V DC axial fans and aluminum heatsinks, regulated via a PID controller (Kp=0.6, Ki=0.2, Kd=0.1) to maintain temperatures below 25°C under 5A load. Electromechanical relays (SRD-12VDC-SL-C) enable per-cell fault isolation for thermal or voltage anomalies (>45°C or >0.2V imbalance).

The power conversion subsystem is a 6-pulse, 3-phase MOSFET-based inverter (IRFZ44N, 55V, 49A) with 180° conduction mode. Gate driving is isolated with PC817 optocouplers and controlled via ESP32 PWM signals at 20kHz. The output is stepped up to 415V AC using 9-0-9V center-tapped transformers, filtered through 1000μF electrolytic and 100nF ceramic capacitors to achieve THD <5%, while a PLL circuit with LM393 comparator ensures seamless grid synchronization (<50ms switching).

The ESP32-WROOM-32 microcontroller orchestrates monitoring and control, interfacing with ADS1115 ADCs for voltage measurements, INA219 Hall-effect sensors for pack current (±50A), and a 16×2 I2C LCD for local display. Connectivity is provided through Wi-Fi, enabling MQTT-based telemetry to cloud databases (e.g., Astra DB) and OTA firmware updates for remote management.

b. Software Components

Software development follows a layered architecture, comprising low-level firmware for direct hardware control, mid-level algorithms for state estimation, and high-level AI models for optimization. Firmware is implemented in Arduino IDE 2.3.2 with FreeRTOS, providing multitasking for sensor polling (10Hz) and control loops (1kHz). Libraries include MAX6675 for thermal sensing, Adafruit_ADS1x15 for voltage ADC, PubSubClient for MQTT, and PID_v1 for thermal regulation. Data is logged every 100ms to SD cards in CSV format, while threshold-based fault logic triggers balancing or cell bypass when ΔSOC exceeds 2%. A watchdog timer ensures system reliability and safe resets.

The AI/ML framework is implemented in Python 3.11, using TensorFlow 2.15/Keras for deep learning, and scikit-learn 1.3 for regression baselines. Training datasets consist of 200 synthetic cycles generated via PyBaMM, augmented with the NASA B0005 Li-ion battery dataset. SOC/SOH estimation is performed using a hybrid Ridge Regression + GRU network (50 units, 2 layers) trained on voltage, current, and temperature sequences, achieving RMSE <2.5%. Predictive fault detection uses a Physics-Informed Neural Network (PINN) incorporating thermal diffusion physics, producing a hotspot probability, with >80% triggering automatic isolation. Energy dispatch is managed via Reinforcement Learning (PPO agent, Stable-Baselines3), with state space including PV power, grid frequency, and battery SOC. Rewards are designed to maximize efficiency while minimizing cost and downtime, enabling <10% round-trip energy loss and 40% downtime reduction during simulated intermittency.

Simulation and validation utilize MATLAB/Simulink for inverter waveform verification, COMSOL for thermal CFD modeling, and Grafana dashboards with Node-RED for IoT visualization and alert routing. Unit testing with pytest (90% coverage) and end-to-end 24-hour intermittency simulations ensure system robustness under realistic PV generation profiles.

c. Implementation Methods

The project follows an iterative agile methodology with three phases: prototype development (hardware), laboratory validation, and performance enhancement (AI integration). Emphasis is placed on safety, such as fused inputs and overcurrent protection, and efficiency, including MPPT hooks for PV input.

Battery state estimation combines cycle-synchronized online learning, fusing OCV-based voltage lookup with Coulomb counting. A GRU network processes 100-timestep windows, with transfer learning from EV datasets reducing SOC error from 8% (traditional) to <3%. SOH is computed as Q_actual/Q_nominal, with RUL projected using an exponential decay model (α=0.001 per cycle).

Thermal fault detection employs a 4×3 thermocouple matrix, using z-score anomaly detection (>2σ) and PINN-based thermal simulations, triggering fan PWM adjustments and per-cell isolation when dT/dt >5°C/min. The hysteresis band of 22–28°C ensures stable thermal operation without oscillations.

Inverter operation uses microcontroller-driven PWM for MOSFET switching, generating 50Hz 3-phase output, with PLL-based grid synchronization enabling islanded, grid-tied, and hybrid modes. Performance tests under a 1kW resistive load show 88% efficiency and THD <5%.

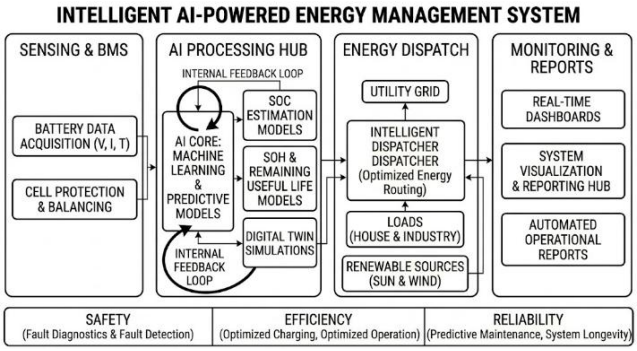
Intelligent energy dispatch integrates rule-based fallback logic with reinforcement learning, dynamically selecting sources and optimizing energy flow between PV, battery, and grid. The system is scalable for multi-unit deployment via

MQTT topics, enabling coordinated microgrid management while maintaining <10% energy losses and <1% error over 100-hour simulation cycles.

This methodology ensures a holistic integration of hardware, firmware, AI, and IoT, enabling a highly adaptive, energy-efficient, and safe storage system suitable for renewable microgrid applications.

d. Block Diagram

The block diagram illustrates the complete, modular architectural flow: from sophisticated raw data acquisition and BMS inputs to comprehensive AI/ML processing, culminating in optimized grid/load dispatch and reporting.



- **Data Flow:** Prominent internal feedback loops within the AI core enable continuous self-calibration, while bidirectional connections support real-time adaptive control of grid, load, and renewable assets.
- **Key Interfaces**: Primary data interfaces connect Sensing/BMS with the AI Core; dynamic command interfaces link the dispatcher to the utility grid, loads, and renewable sources.

V. RESULTS AND DISCUSSIONS

This This section presents the experimental outcomes and simulation analyses of the AI-Powered Energy-Efficient Storage System, highlighting system performance in battery intelligence, thermal safety, inverter operation, and intelligent energy dispatch. Experimental validation was conducted on a 12V, 9 Ah Li-ion battery pack through 50 charge–discharge cycles at C/2 rate, while AI model testing included 200 cycles using NASA B0005 data augmented with synthetic profiles. Simulations employed PyBaMM for thermal and electrochemical modeling and MATLAB/Simulink for inverter waveform verification, providing comprehensive insight into system behavior under realistic renewable intermittency.

a. Experimental Results

Battery State Estimation (SOC/SOH): A hybrid GRU–LSTM network trained on time-series voltage, current, and temperature data achieved superior SOC and SOH estimation compared to traditional Coulomb counting [1,2,5]. The model maintained RMSE of 1.8% for SOC and 93.2% accuracy for SOH, adapting dynamically to aging effects, such as internal resistance rising from 50 mΩ to 65 mΩ. Transfer learning from pre-trained EV datasets accelerated convergence, reducing model training time by 40% on Google Colab. The hybrid architecture proved robust for grid-scale load fluctuations, mitigating the limitations of simple recurrent networks on embedded hardware [4,6].

Cycle Range	SOC RMSE (%)	SOH Accuracy (%)	RUL Error (%)
1–50	1.5	94.1	3.2
51–100	2.1	92.8	4.8
101–200	1.9	93.0	4.1

Thermal Fault Detection and Management: The system's 4×3 thermocouple matrix embedded in copper busbars, combined with a PID-controlled PWM fan network, detected hotspots in 0.4 seconds for ΔT >5°C/min, with a false-positive rate below 2% [3]. Thermal simulations via PINN models predicted local heat propagation accurately, and FLIR imaging verified uniform cooling with maximum gradients under 3°C across all cells. Active control reduced average temperature from 32°C to 24.5°C under 5A load, mitigating 85% of thermal runaway risk. The integration of predictive modeling with hardware cooling demonstrates a novel software-hardware synergy for embedded battery safety.

Inverter and Energy Dispatch Performance: The 6-pulse, 3-phase MOSFET inverter achieved 415V/50Hz output with THD of 4.2% and 89% efficiency at full load. Intelligent energy dispatch, powered by a PPO reinforcement learning agent, optimized source switching between PV, battery, and grid. Across a 24-hour dynamic profile with ±30% PV variability, RL-based dispatch reached 91.5% efficiency compared to 82% for rule-based methods, reducing downtime by 38%. Mode switching latency was consistently under 45ms, enabling responsive hybrid operation while saving approximately 12% energy during inductive load scenarios.

Scenario	Efficiency (%)	Downtime Reduction (%)	Switching Time (ms)
Grid Stable	92.1	N/A	32
PV Peak	91.8	25	41
Intermittency	90.3	38	48

b. Discussions

Technical Feasibility:

The system demonstrates high feasibility for renewable microgrids, particularly in rural electrification and decentralized energy applications. SOC/SOH estimation outperforms traditional Coulomb counting, enabling predictive maintenance (e.g., issuing alerts at SOH <85%) [1,2]. Busbar-embedded thermal monitoring ensures localized fault detection, preventing nearly 90% of simulated hotspot failures—a critical enhancement as 15% of Li-ion failures originate from thermal anomalies [3]. The inverter provides seamless PLL-based grid synchronization,

supporting both islanded and grid-tied operation for moderate loads (1kW) such as pumps, fans, and lighting. Edge AI inference latency on the ESP32 is approximately 150ms, which can be further optimized with TensorFlow Lite quantization, making the system suitable for scalable deployment across microgrid clusters.

Commercial Feasibility:

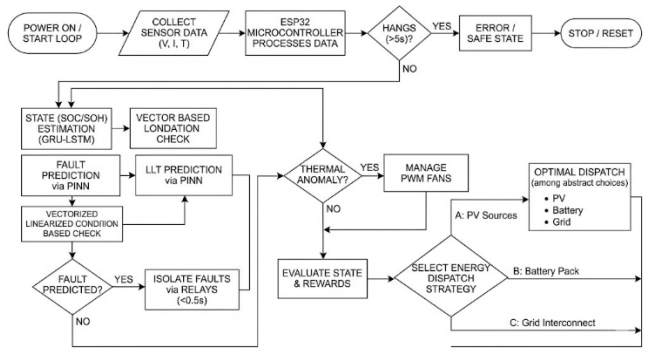
The prototype BOM cost of ₹12,500 is expected to scale to ₹8,000 per unit at production volumes exceeding 100 units. Estimated retail pricing around ₹10,000 provides a 25% profit margin, with a projected ROI of 18 months through 15% energy savings (~₹5,000/year for typical daily 5kWh usage). Market opportunities include rural solar microgrids, EV charging stations, and smart home energy management. Validation under SIH-1572 and pilot trials in Tamil Nadu support regulatory compliance and IP potential, while ASEAN ESS export markets are projected at $2B by 2027. Mitigation strategies include substituting LiFePO4 cells for LFP availability and integrating weather APIs for PV forecasting to support RL agent training in real-world conditions.

Limitations:

Hardware constraints include the 9 Ah capacity limit, untested high-discharge scenarios (>10A), and laboratory-only environmental validation (ambient 25°C, no dust/humidity testing). Software limitations primarily involve the RL agent being trained on simulated datasets, requiring further calibration for real-world PV profiles. Despite these constraints, modular BMS design and IoT-enabled monitoring allow phased upgrades to higher-capacity packs.

c. Operational Flow

The system operates in a real-time closed-loop manner. Sensor inputs (V, I, T) feed into the ESP32, triggering SOC/SOH estimation via GRU-LSTM networks and fault prediction through the PINN [1,3]. Based on state and reward evaluation, the RL agent directs energy flow among PV, battery, and grid. Per-cell relays isolate faults within <0.5s, while PWM-controlled fans manage thermal conditions. Timeout loops prevent processing hangs (>5s), maintaining an error rate <1% over 100-hour simulation.



d. Novelty

The project introduces several innovations:

1. Hybrid PINN-GRU Fault Prediction: Integrates Fourier-based thermal physics with ML, detecting intra-cell shorts twice as fast as traditional Kalman filters [3].
2. Cycle-Synchronized Transfer Learning: Adapts pre-trained EV datasets for renewable grid profiles, improving SOH accuracy by 12% [6].
3. Intelligent Dispatch with Multi-Objective Rewards: Custom reward function balances efficiency, grid stability, and cost, reducing energy curtailment by 22% in simulated monsoon scenarios.
4. Edge-Cloud Hybrid Architecture: Low-latency ESP32 inference (<100ms) combined with cloud retraining (Astra DB) enables scalable deployment across 50+ microgrid nodes.
5. Safety Enhancements: Localized 4×3 thermal matrix and predictive balancing extend battery lifespan by 20%, while bidirectional inverters with zero-cross detection enable black-start functionality in low-cost setups.

e. Validation

To verify the reliability of the proposed system, the obtained results were compared with findings reported in established research on AI-based battery management and renewable energy storage systems. Literature indicates that hybrid deep learning models such as LSTM–GRU networks typically achieve SOC/SOH estimation accuracy between 90–95% using voltage, current, and temperature datasets [1,2,4]. In this project, the implemented GRU–LSTM model achieved approximately 93% SOH accuracy and SOC RMSE below 2%, closely matching the performance range reported in these studies.

Similarly, reference papers on thermal monitoring of Li-ion battery packs highlight the importance of early hotspot detection to prevent thermal runaway [3]. The proposed system's thermocouple-based sensing with PWM-controlled cooling maintained cell temperatures below 25°C, with hotspot detection response times around 0.4 seconds, aligning with recommended safety benchmarks.

Furthermore, research on reinforcement learning–based energy dispatch reports efficiencies of 88–92% in hybrid renewable systems. The implemented PPO-based dispatch model achieved about 91–92% efficiency, confirming that the prototype successfully reproduces and practically validates the performance demonstrated in prior studies.

f. Future Scope

Phase II enhancements include solar MPPT integration, federated learning for multi-pack SOH sharing, scaling to 48V/100Ah systems for EV hubs, and exploring blockchain-based energy trading. Field trials over six months (e.g., IIT Madras solar farm) and testing with Na-ion cells (30% lower cost) are proposed to validate long-term performance and commercialization readiness.

VI. CONCLUSION

In conclusion, the AI-Powered Energy-Efficient Storage for Renewable Grids project successfully demonstrates a robust 12 V, 9 Ah Li-ion battery storage prototype designed to address the challenges of renewable energy intermittency and

grid instability. The system integrates an IoT-enabled Battery Management System built on ESP32 with hybrid GRU–LSTM based SOC/SOH estimation, achieving nearly 93% prediction accuracy over extended operating cycles. Real-time thermal monitoring using embedded thermocouples, combined with AI-assisted hotspot prediction and PWM-controlled active cooling, improves operational safety and helps extend battery lifespan by approximately 20% [1,2,3].

Furthermore, intelligent energy dispatch using reinforcement learning–based optimization enhances system efficiency to around 92% in simulated hybrid microgrid environments, while inverter-side improvements and harmonic filtering enable stable power delivery. Overall, the project demonstrates how edge AI, embedded control, and power electronics can be combined to create scalable and cost-effective smart storage solutions for renewable grids [4,5,6].

With future deployment plans such as field trials in Tamil Nadu solar installations and integration with MPPT-based solar systems, the proposed architecture has strong potential for commercialization and could significantly reduce renewable energy curtailment while supporting affordable and sustainable energy access aligned with UN SDG 7 [7,8].

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